ROBERTA HENRIQUEZ

ELECTRICAL AND COMPUTER ENGINEER
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EDUCATION

University of Memphis | Herff College of Engineering Memphis, TN
Dual bachelor's degrees in Electrical Engineering and Computer Engineering • Expected Graduation: **May 2027**

TECHNICAL SKILLS

Programming: Python, C/C++ (embedded), MIPS Assembly, JavaScript, SQL, MATLAB/Simulink
AI and Machine Learning: TensorFlow, TensorFlow Lite, Transfer Learning (ResNet50), Computer Vision, XGBoost, Random Forest, K-Means/GMM
Hardware and Embedded: ESP32, IMU Sensors (BNO055), FLIR Thermal Imaging, Analog Front-End Design, Oscilloscopes, Power Instrumentation
Software and Web: Django, Next.js, Node.js, SQLite, REST APIs, Git/GitHub | **Regulatory:** FDA 510(k), Software as a Medical Device

ENGINEERING AND ENTREPRENEURIAL EXPERIENCE

ThermalSense | Founder and Lead Engineer Spring 2026 – Present
- Founded an AI-powered thermal imaging platform for early breast cancer screening in rural and underserved communities.
- Fine-tuned a **ResNet50** model on the DMR-IR breast thermography dataset, reaching **87%** validation accuracy and **0.92 AUC**; optimized it for on-device inference with TensorFlow Lite.
- Architected an operator-run screening system around the FLIR ONE Edge Pro thermal camera, with encrypted on-device image storage to protect patient privacy.
- Defined an FDA Class II **510(k)** regulatory pathway with a Software as a Medical Device layer; consulted FDA and IRB advisors on human-subjects data collection.
- Built the B2B pricing model for pharmacies and hospitals, authored the investor pitch deck, and secured full IP ownership through UofM Technology Transfer.

ENGINEERING PROJECTS

AI-Assisted Navigation System | Senior Design Team Lead Fall 2026 – Present
- Lead a **5-member** engineering team with Dr. Madhu Balasubramanian to design a smart sensor cane paired with an AI camera headset that detects obstacles and describes surroundings through audio.
- Direct project scoping, problem definition, and twice-weekly design reviews against a **May 2027** delivery target.

Service Status and Incident Management Platform | Team Lead Fall 2026 – Present
- Lead a **4-member** software team building automated health checks, a 4-stage incident state machine, email and webhook alerts, and public and admin dashboards.
- Own the AI integration: LLM-drafted incident summaries routed through a human-in-the-loop approval workflow.

Good Posture Reminder Device | Junior Design Spring 2026
- Built a wearable using an **ESP32** and **BNO055 IMU** that detects posture deviation in real time and delivers threshold-based haptic feedback.

Workforce Management Platform | Full-Stack Developer Summer 2026
- Built and deployed a production **Next.js** and **SQLite** web app on Vercel for a cross-border engineering staffing company, with separate admin, employer, and candidate portals.
- Developed visa-expiration and consulate-document tracking, resume uploads, secure admin-issued accounts, and a revenue dashboard (MRR/ARR).

LEADERSHIP, HONORS, AND LANGUAGES

Founder | ThermalSense • **Team Lead** | Novatech Senior Design and TheCoders Software Team
Competitions: NIH DEBUT Challenge (2026), AWS AI Pitch Competition, Memphis (2026)
Accelerators: Accepted to Mason I-Corps NextGen Technology Cohort, Mid-South Hub (2026)
Languages: Trilingual in English, Spanish, and Italian